

CSE440: Natural Language Processing II

Lab Project

Multi-Class Text Classification: A Comparative Analysis

Task:

You will be assigned one of the following datasets: You must use only your assigned dataset throughout the project.

Data Access Link: [Project Dataset Spring26](#)

Project Guidelines: [READ THIS VERY CAREFULLY!]

Begin by loading and exploring your assigned dataset. The dataset is already split into training and testing, provided as separate CSV files. Use the training set for model training and validation, and the testing set strictly for final evaluation.

In this project, you will do extensive Exploratory Data Analysis (EDA) to uncover meaningful insights from your dataset. Report your findings in both the project report and the corresponding .ipynb notebook. After EDA, you will experiment with different pre-processing techniques that you learnt throughout the course to get clean text. You will make three preprocessed data:

1. A dataset with no preprocessing:

Here, you will not use any preprocessing techniques and before converting the text into tf-idf vectors or other embedding techniques.

2. A dataset with extreme preprocessing:

Here you will do all the preprocessing you learnt throughout the course including stemming.

3. A dataset with optimum preprocessing:

Based on your EDA, you will decide which preprocessing needs to be done.

Now, you will implement various word representation techniques and apply different Machine Learning (ML) and Neural Network (NN) models. You are required to implement all the word representation techniques covered in the labs and assignments, which include:

- TF-IDF (For ML and Deep NN Models)
- Skip-gram (For all NN models)

Note that, for skipgram, you can either train the representation yourself using the given dataset or you can download a pretrained one.

Next, you will train any **one** of the machine learning models below:

- Random Forest
- Logistic Regression
- Naive Bayes

Then, you will train **all** of the following neural network models below:

- Deep Neural Network (Should have at least 3 to 4+ layers and 128+ Neurons in the first layer) (TF-IDF)
- SimpleRNN (Skipgram)
- GRU (Skipgram)
- LSTM (Skipgram)
- Bidirectional SimpleRNN (Skipgram)
- Bidirectional GRU (Skipgram)
- Bidirectional LSTM (Skipgram)
- BERT Base (Optional)

You are not required to implement Transformer-based or other cutting-edge models, but you are welcome to explore them if time permits.

For each model, you must manually tune the hyperparameters, using validation performance to guide your choices. For each preprocessed dataset, you are required to experiment with:

- TF-IDF with one ML model and a deep neural network.
- Skip-gram with all neural network models.

You need to tune the hyper parameters while experimenting with the models.

Note that the experiments will require at least 20 to 30 separate training runs. Therefore, start your project early. Divide the work with your groupmates. You will not be able to finish it if you begin on the last day. Also, usage of AI is allowed as long as you understand what you are doing.

Evaluate the performance of each model (paired with the word representation techniques) using appropriate metrics. You must calculate and report the following:

- Accuracy

- F1-score (macro)
- Confusion matrix
- Classification report

Use both visual and tabular representations to compare model performance clearly and effectively.

At the end of your experiments, you must:

- Identify and highlight the best-performing and worst-performing models and preprocessing techniques.
- Provide a detailed comparison among all of your experiments.
- Your report should contain the reasons for all decisions made including choosing the hyperparameter of each model.

Deliverables:

Your final submission must include the following components:

Jupyter Notebook (.ipynb File)

- The notebook must be well-organized and clearly structured.
- Use appropriate Markdown cells for section headings, explanations, and analysis.
- Include comments/titles above code blocks to ensure readability and clarity. **However, do not add comments to individual lines, otherwise you will be instructed to remove those before the viva.**
- All plots, tables, and evaluation results should be embedded within the notebook.
- Clearly indicate any hyperparameter tuning experiments and log the results accordingly.
- Make sure your code has output logs shown. Without output logs, we won't be able to verify your codes.

Project Report (IEEE Double-Column Format, Up to 6 Pages – PDF)

Latex Word

- **Abstract:** A brief summary of your overall approach, techniques used, and key findings.
- **Introduction:** Background of the task, motivation for the project, and a brief overview of the dataset.
- **Methodology:**
 - Dataset: Description of the dataset and your findings on EDA. Should contain graphs, word clouds and other visualizations.
 - Preprocessing: Details of all the preprocessing you performed and the word representation techniques
 - Models: Details of your model architectures.

- **Results:** Comparative analysis of the models using relevant evaluation metrics. Include tables and visualizations to support your discussion.
- **Conclusion:** Summarize the key takeaways from your experiments, note any limitations encountered, and suggest possible future improvements.
- **References:** Cite all research papers, tools, libraries, or resources used in your project. Use IEEE citation style.

Viva/Presentation:

- You will attend a short viva or presentation session where you will explain your approach, design decisions, and the results you obtained.
- Each team member must be prepared to answer questions about any aspect of the project, including the code and methodologies. It does not matter who did what, you must be able to answer all the questions.
- As the use of AI tools is permitted, you are expected to have a clear understanding of your implementation and be able to explain your work confidently. **You may be asked specific questions about parts of your code where AI was used or where concepts were applied that were not covered in class. If you are unable to explain these sections, you will be penalized.**

Submission Instructions:

You must submit a compressed ZIP file containing the following:

- Your report (GroupNo_ID1_ID2_ID3.pdf)
- All code files or notebooks (GroupNo_ID1_ID2_ID3.ipnyb/py)

Mark Distribution

The final grading will be based on the following components:

- Viva/Presentation: 7 marks
- Code: 2 marks
- Report: 2 marks